

CSCI E-102 Econometrics and Causal Inference with R

Harvard Extension School

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Lecture 11

- 1 Censored Regression Models
 - Examples of Censored Data
 - Tobit Model
 - Assumptions
 - Likelihood for Tobit Model
- 2 Difference in Differences (DD)
 - Idealized Experiment: Difference Estimator
 - Quasi-Experiment: DD Estimator
 - Example
 - DD Estimator
- 3 Regression Discontinuity Design (RDD)
 - Sharp RDD
 - Fuzzy RDD

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Examples of Censored Data

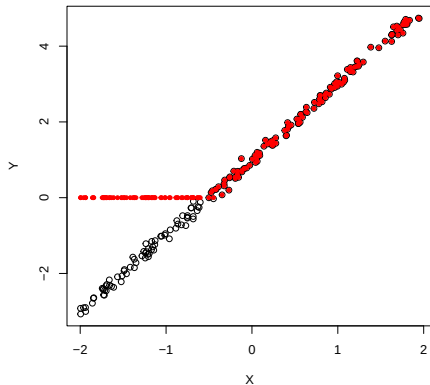
Example:

```
n = 200
X = runif(n, min = -2, max = 2)
u = rnorm(n, sd = 0.1)
Y_latent = 1 + 2*X + u
Y = Y_latent; Y[Y<0] = 0

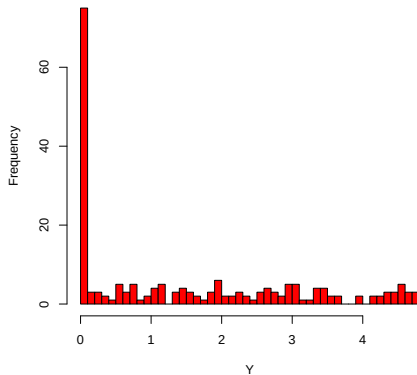
par(mfrow=c(1,2))
plot(X, Y_latent, pch = 1, xlab = "X", ylab = "Y")
lines(X,Y, col = "red", type = "p", pch = 20)
hist(Y, 40, col="red")
```

Examples of Censored Data

Example:



Histogram of Y



Examples of Censored Data

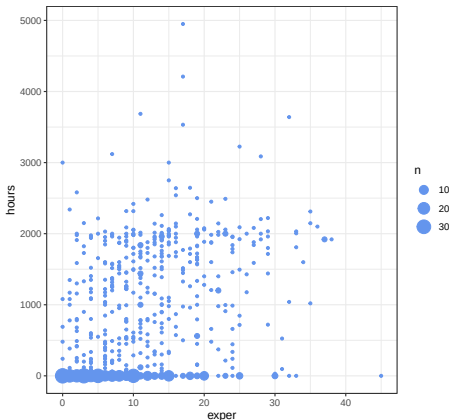
Example:

```
ggplot(mroz, aes(exper, hours)) +  
  geom_count(colour="cornflowerblue") +  
  theme_bw() +  
  ggtitle("Data (1975) on hours worked and labor market experience  
          \nfor 753 married women")  
  
hist(mroz$hours, 40,  
     col="cornflowerblue",  
     xlab="hours",  
     main="Histogram of hours")
```

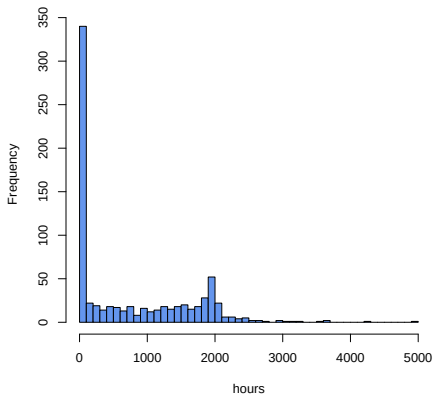
Examples of Censored Data

Example:

Data (1975) on hours worked and labor market experience for 753 married women



Histogram of hours



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Censored Regression Models: Assumptions of Tobit

Tobit Model:

$$Y_i = \begin{cases} Y_i^*, & \text{if } Y_i^* \geq 0, \\ 0, & \text{if } Y_i^* < 0, \end{cases}$$

where *latent* variable $\mathbf{Y}^* = (Y_1^* \ Y_2^* \ \dots \ Y_n^*)^\top$ is given by:

$$\mathbf{Y}^* = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon},$$

where

$$\mathbf{Y} = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix}, \quad \boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} 1 & X_{11} & X_{12} & \dots & X_{1k} \\ 1 & X_{21} & X_{22} & \dots & X_{2k} \\ \vdots & & & \ddots & \\ 1 & X_{n1} & X_{n2} & \dots & X_{nk} \end{bmatrix}, \quad \text{and } \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix}.$$

Typically, we assume $\varepsilon_i \stackrel{\text{iid}}{\sim} \text{Normal}(0, \sigma^2)$, i.e. normality and homoskedasticity.

Censored Regression Models: Likelihood for Tobit

Claim:

Let's consider a Tobit model, where we assume normality and homoskedasticity of errors:

$$\varepsilon_i \stackrel{\text{iid}}{\sim} \text{Normal}(0, \sigma^2), \quad i \in \{1, 2, \dots, n\},$$

then for the log-likelihood we have:

$$\ell(\beta, \sigma | \underbrace{\mathbf{X}, \mathbf{Y}}_{\text{data}}) = \sum_{Y_i > 0} \left[-\frac{1}{2} \ln(2\pi) - \frac{1}{2} \ln(\sigma^2) - \frac{\varepsilon_i^2}{2\sigma^2} \right] + \sum_{Y_i = 0} \ln \left[1 - \Phi \left(\frac{\mathbf{X}\beta}{\sigma} \right) \right].$$

Here,

$$\Phi(z) \doteq \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt$$

denotes the cdf of a standard normal random variable, as always.

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Idealized Experiment: Difference Estimator

Let

- Y_i be an outcome variable and
- X_i be a binary treatment indicator, that is,

$$X_i = \begin{cases} 1, & \text{if } i \text{ is in treatment group,} \\ 0, & \text{if } i \text{ is in control group.} \end{cases}$$

Difference Estimator:

$$Y_i = \beta_0 + \beta_1 X_i + u_i, \quad i = 1, 2, \dots, n.$$

If X_i is randomly assigned, $E[u_i|X_i] = 0$, i.e. OLS is unbiased and consistent.

Idealized Experiment: Difference Estimator

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Difference Estimator with Additional Regressors:

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 W_{i1} + \dots + \beta_{r+1} W_{ir} + u_i, \quad i = 1, 2, \dots, n.$$

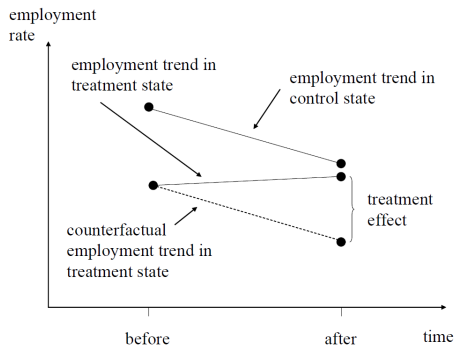
Note that including W 's reduces the standard error of the regression.

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Quasi-Experiment: Example

Average employment per store before and after the New Jersey minimum wage increase

Variable	PA (i)	NJ (ii)	Difference, NJ-PA (iii)
1. FTE employment before, all available observations	23.33 (1.35)	20.44 (0.51)	-2.89 (1.44)
2. FTE employment after, all available observations	21.17 (0.94)	21.03 (0.52)	-0.14 (1.07)
3. Change in mean FTE employment	-2.16 (1.25)	0.59 (0.54)	2.76 (1.36)



Source: *Mostly Harmless Econometrics* by J. Angrist and J. Pischke

Quasi-Experiment: DD Estimator

Let

- Y_{it} be an outcome variable for entity i at time $t = 1, 2$ and $\Delta Y_i \doteq Y_{i2} - Y_{i1}$
- X_i be a binary treatment indicator, that is,

$$X_i = \begin{cases} 1, & \text{if } i \text{ is in treatment group,} \\ 0, & \text{if } i \text{ is in control group.} \end{cases}$$

DD Estimator:

$$\Delta Y_i = \beta_0 + \beta_1 X_i + u_i, \quad i = 1, 2, \dots, n.$$

Quasi-Experiment: DD Estimator

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DD Estimator:

$$\Delta Y_i = \beta_0 + \beta_1 X_i + u_i, \quad i = 1, 2, \dots, n.$$

DD with Additional Regressors:

$$\Delta Y_i = \beta_0 + \beta_1 X_i + \beta_2 W_{i1} + \dots + \beta_{r+1} W_{ir} + u_i, \quad i = 1, 2, \dots, n.$$

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Regression Discontinuity Design (RDD): Sharp RDD

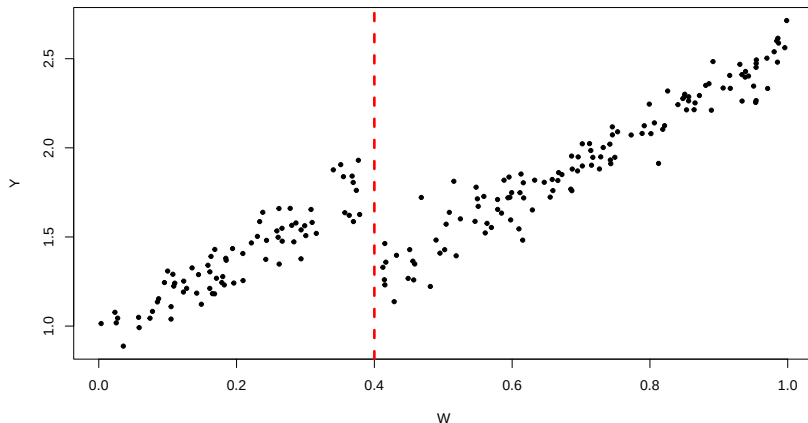
Example:

```
n=200
W = runif(n)
X = 1*(W > 0.4)
u = rnorm(n, sd = 0.1)
Y = 1 - 0.5*X + 2*W + u

plot(W, Y, pch = 20, xlab = "X", ylab = "Y")
abline(v = 0.4, lty=2, lwd=3, col = "red")
```

Regression Discontinuity Design (RDD): Sharp RDD

Example:



Regression Discontinuity Design (RDD): Sharp RDD

Sharp RDD:

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 W_i + u_i, \quad i = 1, 2, \dots, n,$$

where

$$X_i = \begin{cases} 1, & \text{if } W_i < w_0, \\ 0, & \text{if } W_i \geq w_0 \end{cases}$$

for some known w_0 .

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Regression Discontinuity Design (RDD): Fuzzy RDD

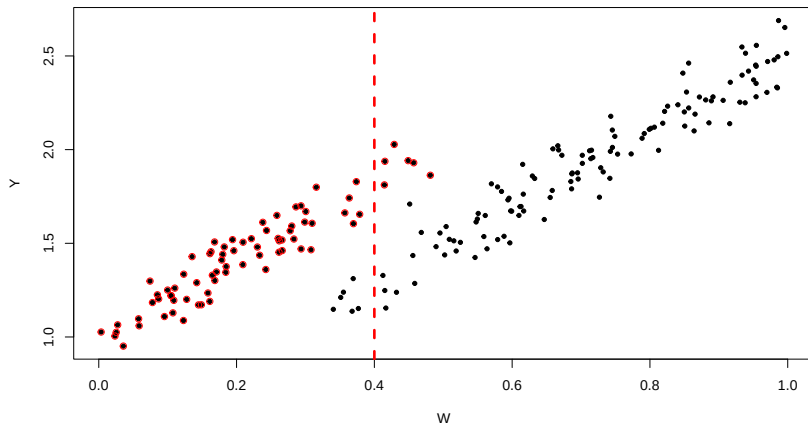
Example:

```
n=200
W = runif(n)
X = 1*(W + rnorm(n, sd=0.05) > 0.4)
u = rnorm(n, sd = 0.1)
Y = 1 - 0.5*X + 2*W + u

plot(W, Y, pch = 20, xlab = "W", ylab = "Y")
abline(v = 0.4, lty=2, lwd=3, col = "red")
lines(W[X<0.4],Y[X<0.4], type = "p", col = "red")
```

Regression Discontinuity Design (RDD): Fuzzy RDD

Example:



Regression Discontinuity Design (RDD): Fuzzy RDD

Fuzzy RDD:

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 W_i + u_i, \quad i = 1, 2, \dots, n,$$

via 2SLS, where

- X_i is a binary variable which indicates which group i belongs and
- instrumental variable

$$Z_i = \begin{cases} 1, & \text{if } W_i < w_0, \\ 0, & \text{if } W_i \geq w_0 \end{cases}$$

for some known w_0 .